DOD-T-17412C(SH)
20 September 1978
SUPERSEDING
MIL-T-17412B(SHIPS)
20 August 1973
(See 6.4)

METRIC

MILITARY SPECIFICATION

TEST KIT, WATER, ENGINE COOLING SYSTEMS (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a test kit suitable for determining chromate and chloride ion content, and pH value of engine cooling water.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY
MIL-P-116 - Preservation-Packaging, Methods of.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking For Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents from a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, G F Carl, Tarriff Publications Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INCORPORATED, AGENT
National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Incorporated, Agent, 1616 P Street, N.W., Washington, DC 20036.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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AMERICAN CHEMICAL SOCIETY PUBLICATION REAGENT CHEMICALS
ACS Specifications

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street, N.W., Washington, DC 20006.)

(Technical society and technical association specifications and standards are generally available for references from libraries. They are also distributed among technical groups and using Federal Agencies.)

NATIONAL BUREAU OF STANDARDS
NBSIR 74-761 - Calibration of Small Volumetric Laboratory Glassware.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 General. The testing kit shall be suitable for determining chromate and chloride ion concentrations, and pH values of engine cooling water. The kit shall consist of a durable case containing in a convenient, compact manner a comparator with color standards for chromate and pH determinations, dropper bottle containing pH indicator, bottle of chloride test tablets, sampling tube and rubber stoppers, funnel, and filter paper.

3.1.1 Recovered materials. Unless otherwise specified herein, all equipment, materials and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification, unless otherwise specifically specified.

3.2 Case. The case shall be fabricated of wood, plastic, or metal, and shall be provided with a hinged fitted cover and a positive fastener which securely fastens the cover to the case. The case shall be approximately 100 mm by 100 mm by 200 mm in size. The case shall rest securely on a flat surface.

3.3 Color comparator. The color comparator shall include minima and maxima standards for pH and chromate determinations, situated on either side of a sample viewing cell. The solid transparent standards shall be fabricated of sheets of colored transparent material. The comparator may be a separate unit or an integral part of the case, and the viewing cell may be a separate unit or an integral part of the comparator. The comparator, if separate from the case, shall resist upsetting or tipping. The maximum and minimum color standards for the pH determination shall represent limits of 8.25 ± 0.25 and 9.75 ± 0.25 pH units when tested as specified in 4.8.1. The maximum and minimum color standards for the chromate determination shall represent limits of 700 ± 100 and 1700 ± 200 parts per million (p/m) chromate ion ($CrO_4^{=}$) when tested as specified in 4.8.2. The design and dimensions shall be as shown on figure 1.

3.4 Viewing cell. The viewing cell, whether a separate unit or integral part of the comparator shall have a maximum viewing path of 19 mm, or less. When the test kit has separate sample tubes for use in the viewing cell, a total of three such tubes shall be provided to allow for breakage.

3.5 pH indicator. The test kit shall contain one bottle with dropper cap containing 25 milliliters (mL) of pH indicator solution. The bottle shall be made of an amber tinting or low actinic glass, suitable for protecting the light sensitive solution. A liter (L) (1000 mL) of indicator solution shall contain 0.1575 grams (g) of ACS grade phenolphthalein ($C_{20}H_{16}O_4$), 0.0100 g of the purest reagent grade cresol red ($C_{21}H_{18}O_5S$) and 0.2625 g of ACS grade thymol blue ($C_{27}H_{30}O_5S$). The indicator dyes shall be weighed accurately and diluted with 525 mL of pure 95 percent (190 proof) ethyl alcohol (C_2H_5OH). A 1 Normal (1N) potassium hydroxide solution shall be added dropwise until the color is green or blue. The indicator solution shall be mixed thoroughly and allowed to stand until all the dyes are completely dissolved. The indicator solution shall be diluted to 1 L with distilled water. The indicator solution shall then be adjusted to a green color by adding 1N potassium hydroxide dropwise if the solution is yellow, or 1N hydrochloric acid dropwise, if the solution is blue. When tested in accordance with 4.8.3, the indicator shall produce a color change from yellow to purple at approximately 8.0 pH value and purple to distinct purple at approximately 9.7 pH value.

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3.6 Chloride test tablets. The test kit shall contain one amber tinted or low actinic glass bottle containing 100 chloride test tablets. The tablets shall not break on normal handling, but shall dissolve readily in water within the pH and chromate limits specified on the comparator. Each tablet shall contain sufficient ACS grade silver nitrate to precipitate the chloride ion (Cl) present in 50 mL of water containing 100 p/m of chloride ion when tested as specified in 4.8.4.

3.7 Sampling tube and rubber stopper. The test kit shall be provided with a plastic or glass sampling tube and a rubber stopper.

3.7.1 Sampling tube. The sampling tube shall be sufficiently stable on a flat surface to minimize tipping and shall have a single graduation line to indicate a delivery volume of 50 ± 1 mL.

3.7.1.1 Sampling tube accuracy. When tested as specified in 4.8.5, the tube shall contain 50 ± 1 mL.

3.7.2 Rubber stopper. The rubber stopper shall be of a size suitable to securely stopper the sampling tube.

3.8 Funnel and filter paper. The test kit shall contain a plastic or glass funnel of about 50 millimeter (mm) diameter and 100 125-mm diameter fine filter papers (42 Whatman, or equal) in a container.

3.9 Instruction for use. Complete instructions for use shall be durably affixed to the case, preferably on the inside cover, or enclosed within the case and shall be either printed with waterproof ink on waterproof material, or shall be protected from damage by water by means of a transparent film or coating. The instructions shall include the following information.

"INSTRUCTIONS FOR USE

"GENERAL:

"The tests for chromate, pH, and chloride should be conducted using the compact test kit provided for this purpose. This test kit has been specifically designed to enable shipboard personnel to conduct the necessary control tests quickly. The kit contains all the equipment and chemicals necessary to conduct the chromate, pH and chloride tests. The tests for chromate and pH are of the color comparison type in which the color of the test sample is compared with the color of two solid transparent discs representing the specified maximum and minimum limits. The chloride test is of a different type and is so designed that the chloride concentration is determined by noting the color and condition of the sample after the addition of a chloride test table.

"PROCEDURES

"1. Sampling of Cooling Water. Allow water to flush from the drain cock for several seconds before collecting a sample. Cool the sample to at least 80°F before test. If suspended material is present in the sample, it must be removed by filtering sample. Filtering is necessary since the suspended material may interfere with the colorimetric tests.

"2. Filtering of the Water Sample. When suspended matter is present in the sample of cooling water, the sample should be filtered before test. To do this, fold a circular filter paper in half and then fold in half again. Open the folded filter paper so as to form a cone. Place the cone in the funnel provided in the kit. Wet the paper with distilled water and press it to the funnel. Carefully pour the sample water on the filter paper in the funnel and thence into a transparent glass bottle or beaker. Do not allow the level of the water in the funnel to reach the upper edge of the paper. This will prevent the water and suspended matter from passing between the paper and funnel.

"3. Chromate Concentration.

a. Fill a test tube with cooling water sample and place in the center position in the comparator.
b. Hold the comparator to a suitable light source (see note following instructions).

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c. If the color through the sample is between the color of the discs marked 700-1700, the chromate concentration is satisfactory. Be careful to note the color rather than the intensity of color. Do not compare the darkness or lightness of the colors; rather, determine whether the sample is yellower or bluer than the color disc standards.
d. If the color is bluer (bluish-green) than the color disc marked 700, the chromate content is low.
e. If the color is yellower (yellow-green) than the color disc marked 1700, the chromate content is high. High chromate content is not harmful but wasteful.

"4. pH Test.

a. Fill three test tubes to the mark with cooling water sample.
b. Add fifteen drops of pH indicator to one test tube, shake, and place in center position in comparator. Place the other two test tubes in the remaining positions of the comparator.
c. Hold the comparator to a suitable light source (see note following instructions).
d. If the color through the sample is between the color discs marked 8.25 and 9.75, the pH is satisfactory.
e. If the color is yellower (yellowish-green) than the color disc marked 8.25, the pH is low.
f. If the color is bluer (bluish-green) than the color disc marked 9.75, the pH is high.

"5. Chloride Concentration. It is necessary that the sample to be tested contain chromate and be alkaline in order to conduct the chloride test. The chloride test, therefore, should be run after the chromate and pH have been determined to assure the proper conditions. Should it be desirable to perform the chloride test before adjusting the chromate content and pH, the sample must have the characteristic yellow color of chromate.

a. Fill the cylindrical sample bottle to the 50 milliliter mark with cooling water sample.
b. Add one chloride test tablet, stopper and shake until completely dissolved.
c. If the sample develops a brick red color, the chloride content is below the safe maximum limit.
d. If the sample becomes a cloudy yellow-green, the chloride content is high.

NOTE: Use only natural daylight or a daylight fluorescent lamp. Stock Number NSN 9G 6240-427-7448 covers an 8 watt 12 inch daylight fluorescent lamp suitable for this purpose. Do not use incandescent lighting or other light source as the results will be erroneous. When tests are made, the tester should be held up to the brightest part of the sky, but about 30 degrees from the sun. Avoid strong light on the observer's side of the tester as reflections will make the necessary comparisons difficult. If possible, the observer and comparator should be in a shadow."

3.10 Workmanship. The test kits shall have no missing, broken or damaged parts and shall be free from sharp edges which might be injurious to using personnel. The finish of case shall be uniform and free from scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First Article inspection (see 4.3)
(b) Quality Conformance inspection (see 4.4)

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4.3 First article inspection. The test kits shall be subjected to the tests shown in table I.

TABLE I. First Article and quality conformance inspection.

Test	Requirement paragraph	Test paragraph	First article inspection	Quality Conformance inspection	
				Test	Inspection Level (MIL-STD-105) (See 4.4.2)
Visual examination		4.5	x		
Alkalinity	3.3	4.8.1	x	x	S-2
Chromate	3.3	4.8.2	x	x	S-2
pH indicator	3.5	4.8.3	x	x	S-1
Chloride	3.6	4.8.4	x	x	S-2
Accuracy (sampling tube)	3.7.1.1	4.8.5	x	x	S-2

4.4 Quality conformance inspection.

4.4.1 Inspection lot. For purposes of inspection, a lot shall consist of all test kits to be delivered at one time.

4.4.2 Sampling for visual examination. A random sample of completed test kits shall be selected from each lot in accordance with inspection level II of MIL-STD-105 with an AQL level for major and minor defects as shown in table II. The sample units shall be subjected to the visual examination specified in 4.5.

TABLE II. Sampling for visual examination.

For visual examination	Inspection level	AQL (defects per hundred units)
Major defects	II	1.0
Minor defects	II	2.5

4.4.3 Sampling for quality conformance tests. From each lot, sample test kits shall be selected in accordance with MIL-STD-105 for the quality conformance inspection of 4.6. The inspection level shall be as shown in table I.

4.5 Visual examination. The sample test kits selected in accordance with 4.4.2 shall be subjected to a visual examination to verify compliance with all requirements of this specification not involving tests. Classification of defects shall be as shown in table III. Any complete test kit or part in the sample containing one or more defects shall be cause for rejection of the defective kit or part, and if the number of defective test kits or parts in any sample exceeds the acceptance number for the sample, this shall cause rejection of the lot represented by the sample.

TABLE III. Classificaiton of defects.

Critical:

None defined

Major:

101 Parts missing.
102 Part broken or damaged.
103 Fastening device, does not secure cover to case.
104 Cover does not fit case.
105 Instructions not waterproof.
106 Solution bottle not tightly closed.
107 Chloride test tablets broken.
108 Sampling tube not stable on flat surface.
109 Graduation line of sampling tube missing.
110 Not free of sharp edges which might be injurious to using personnel
111 Parts not properly secured within case.

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TABLE III. Classification of defects. - Continued

Minor:	
201	Finish not uniform, not free of scratches.
202	Case does not rest securely on flat surface.

4.6 Quality conformance inspection. The sample test kits selected in accordance with 4.4.3 shall be subjected to the quality conformance inspection of table I. If any sample fails to conform to any quality conformance test, the lot represented by the sample shall be rejected.

4.7 Reagents.

4.7.1 Standardized chromate solution (10,000 p/m $CrO_4^=$).

4.7.1.1 Reagents.

4.7.1.1.1 Potassium iodide solution (10 percent). Ten grams of ACS grade potassium iodide crystals (KI) shall be dissolved in 100 mL of distilled water.

4.7.1.1.2 Sodium thiosulfate solution, standard. Forty-four grams of ACS grade sodium thiosulfate crystals ($Na_2S_2O_3.5\ H_2O$) shall be transferred to a clean 1-L volumetric flask, and distilled water shall be added. The flask shall be stoppered and then shake until crystals are dissolved. The solution shall be diluted to mark with distilled water and standardized against resublimed iodine crystals, or Bureau of Standards potassium dichromate or potassium iodide.

4.7.1.1.3 Starch solution. About 50 mL of distilled water shall be placed in a 150-mL beaker and brought to a boil. About a half-gram of soluble starch shall be sprinkled into the boiling water. The boiling solution shall be stirred for about 1 minute and then allowed to cool.

4.7.1.1.3.1 Procedure. Approximately 25 g of anhydrous reagent grade sodium chromate shall be dissolved in 900 mL of distilled water in a 1000-mL volumetric flask and diluted to the mark with distilled water. To determine the chromate ion concentration of this solution, 5 mL of sample shall be transferred by pipette to a clean casserole and diluted with 45 mL of distilled water. Five mL of potassium iodide solution and 15 drops of diluted sulfuric acid solution (three concentrated H_2SO_4: one water) shall be added and shall be placed in the dark and allowed to stand for 5 to 10 minutes. The solution shall be stirred and about 1 mL of starch solution shall be added and titrated with the standard sodium thiosulfate solution to the disappearance of the blue color. The number of mL of sodium thiosulfate solution used shall be noted. To prepare the standard sodium chromate solution containing 10,000 p/m $CrO_4^=$, A mL of solution shall be placed in a 1000-mL volumetric flask and diluted to the mark with distilled water. The value of A shall be calculated as follows:

$$A = \frac{1,295}{\text{mL} \times \text{Normality (N) of thiosulfate solution used}}$$

4.7.2 Sodium hydroxide solution (0.1N). A 4.0 g of sodium hydroxide, reagent shall be dissolved in 900 mL of distilled water in a 1000-mL volumetric flask and diluted to the mark with distilled water.

4.7.3 Sulfuric acid solution (0.1N). Thirty mL of reagent grade concentrated sulfuric acid (H_2SO_4, d 1.84) shall be slowly poured into approximately 100 mL of distilled water, cooled, mixed thoroughly, and diluted to 1 L in a volumetric flask. One-hundred mL of this solution shall be transferred by pipette and shall be added to 300 mL of distilled water in a 1000 mL volumetric flask and diluted to the mark with distilled water.

4.8 Test procedures.

4.8.1 Alkalinity. A solution containing 1200 p/m chromate ion shall be prepared by pipetting 120 mL of standard chromate solution into a 1000-mL volumetric flask and diluting with distilled water to the mark. The solution shall be divided into four portions. These portions shall be adjusted to pH values of 8.00, 8.50, 9.50, and 10.00 by the additions of

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0.1N sodium hydroxide solution and 0.1N sulfuric acid. The pH shall be determined by means of a glass electrode pH meter. The pH indicator (see 3.5) shall be added as directed to each of the solutions and the colors of the four solutions shall be compared with those of the minimum and maximum color standards by means of the comparator provided with the kit. The color of the minimum color standard shall fall within the range of colors of the test solutions having pH values of 8.00 and 8.50. The color of the maximum color standard shall fall within the range of colors of the test solutions having pH values of 9.50 and 10.00.

4.8.2 Chromate.

4.8.2.1 Minimum color standard. A test solution containing 600 p/m chromate ion shall be prepared by adding 60 mL of standard chromate solution, by a burette, to 920 mL of distilled water and adjusting the alkalinity to a pH reading of 8.00 by the addition of 0.1N sodium hydroxide and 0.1N sulfuric acid. The pH shall be determined by means of a glass electrode pH meter. The solution shall be diluted 1 L with distilled water. A second test solution containing 800 p/m chromate ion shall be prepared by adding, by burette, 80 mL of standard chromate solution to 900 mL of distilled water and adjusting the alkalinity to a pH reading of 8.50 by the addition of 0.1N sodium hydroxide and 0.1N sulfuric acid, and diluting 1 L. The colors of these solutions and the minimum color standard shall be compared by means of the comparator provided with the kit. The color of minimum solid transparent standard shall fall within the range of colors of the test solutions.

4.8.2.2 Maximum color standard. A test solution containing 1500 p/m chromate ion shall be prepared by adding, by burette, 150 mL of standard chromate solution to 825 mL of distilled water and adjusting the alkalinity to a pH of 9.50 by the addition of 0.1N sodium hydroxide and 0.1N sulfuric acid. The solution shall be diluted to 1 L with distilled water. The pH shall be determined by means of a glass electrode pH meter. A second test solution containing 1900 p/m chromate ion shall be prepared by adding, by burette, 190 mL of standard chromate solution to 800 mL of distilled water and adjusting the alkalinity to a pH of 10.0 by the addition of 0.1N sodium hydroxide and 0.1N sulfuric acid. The solutions shall be diluted to 1 L with distilled water. The colors of these solutions and the maximum color standard shall be compared by means of the comparator provided with the kit. The color of the maximum solid transparent color standard shall fall within the range of colors of the test solutions.

4.8.3 pH indicator.

4.8.3.1 Standard solutions. Two solutions of reagent grade monosodium phosphate ($NaH_2PO_4H_2O$) containing 5 and 50 p/m of phosphate, and a 0.02 normal solution of sodium hydroxide shall be prepared.

4.8.3.2 Procedure. Fifty mL of one of the standard phosphate solutions shall be transferred to a clean dry beaker. Eight drops of the indicator solution shall be added and titrated with the sodium hydroxide solution to disappearance of yellow and the appearance of a distinct purple color. The pH at both color changes shall be determined by glass electrode, pH meter equipment. A similar titration and pH determination shall be made on the second standard phosphate buffer.

4.8.4 Chloride. A test solution containing 80 p/m chloride ion and 700 p/m chromate ion shall be prepared by dissolving 0.131 gram sodium chloride, reagent grade in 500 mL distilled water, adding with a burette 70 mL of standard chromate solution, and diluting to the mark in a 1000-mL volumetric flask. A second test solution containing 700 p/m chromate ion and 120 p/m chloride ion shall be prepared by dissolving 0.197g sodium chloride in 500 mL of distilled water, adding with a burette 70 mL of standard chromate solution, and diluting with distilled water to the mark in a 1000-mL volumetric flask. Each of two sampling tubes (see 3.7) shall be filled to the mark with the above solutions, and a chloride tablet (see 3.6) shall be dropped in each stopper and shake. The speed of disintegration of the tablet and resultant color shall be noted. The water containing 80 p/m chloride ion shall develop a brick red color. The water containing 120 p/m chloride ion shall remain yellow.

4.8.5 Sampling tube accuracy. Accuracy of graduation on sampling tube shall be tested as specified in National Bureau of Standards NBSIR 74-761 SD Catalog No. C13.

4.9 Inspection reports. First article and quality conformance inspection reports shall be furnished in accordance with the data ordering documents specified (see 6.2.2).

4.10 Inspection of preparation for delivery. Packaging, packing, and marking shall be examined to determine conformance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government contracts. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation - packaging. Preservation - packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A. Test kits shall be packaged in accordance with method III of MIL-P-116 as follows: Components of each test kit, shall be contained within the case (see 3.2) by the application of interior blocking, individual wrappings, molded forms, partitions or other such media that will insure the integrity of each component from breakage or loss of contents. Each test kit case shall be further cushioned and individually packaged in a fiberboard box conforming to PPP-B-636 class domestic. Fiberboard boxes shall be closed method I in accordance with the appendix to the box specification.

5.1.2 Level C. The test kits shall be individually preserved - packaged in such a manner that will afford protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. The contractor's commerical practice may be used when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Test kits, packaged as specified (see 6.2.1), shall be packed in a fiberboard box conforming to PPP-B-636, type SF, class weather resistant, grade V, with class and style at the contractor's option. Boxes shall be closed and reinforced in accordance with method V of the appendix to the box specification. The gross weight of the boxes shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Test kits, packaged as specified (see 6.2.1) shall be packed in a fiberboard box conforming to PPP-B-636, class domestic and of minimum 275 grade, normal requirements with variety and style at the contractor's option.

5.2.3 Level C. Test kits, packaged as specified (see 6.2.1) shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to the special marking required herein or by the contract, unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each bottle of pH indicator solution and chloride test tablets shall be labeled and the following markings included thereon.

National stock number (or manufacturers part number if no NSN available).
Manufacturer's name.
Quantity.
Lot number.
Date of manufacture, month, and year.

6. NOTES

6.1 Intended use. The test kit covered by this specification is intended for use in determining chromate and chloride ion content, and pH value of engine cooling water.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable levels of packaging and packing required (see 5.1 and 5.2).

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6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
4.9	Inspection reports	UDI-T-23473	

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

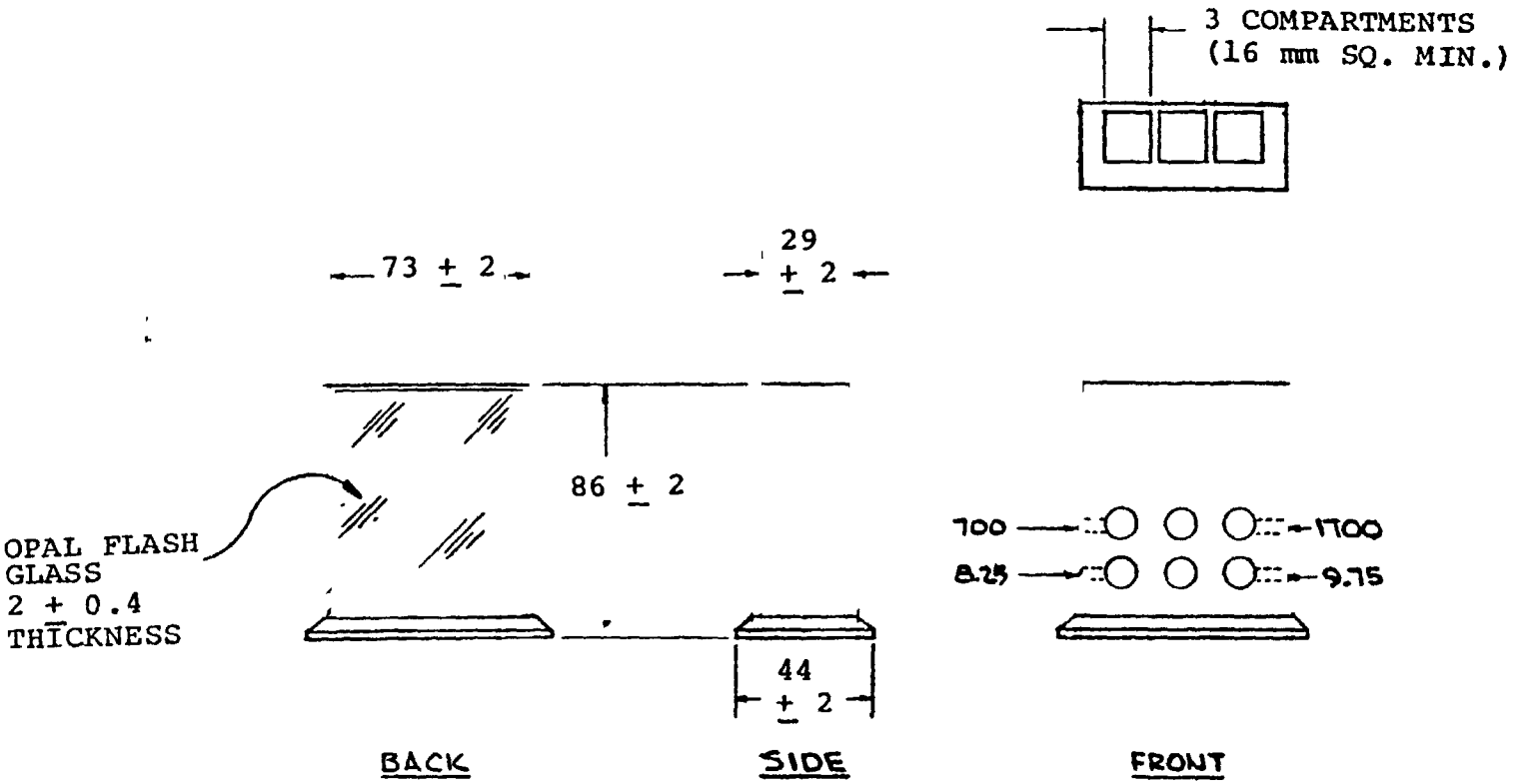
6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness the of the changes.

6.5 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Preparing activity:
Navy - SH
(Project 6630-N310)

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FIGURE 1. Comparator.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE. This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

DoD-T-17412C

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